Proposed Capstone Project - Software Prototype

Title: Intelligent Academic Workload Planner Using Canvas Data and AI-Based Time Estimation

Introduction:

For my capstone project I am proposing a software prototype that integrates with canvas to automatically retrieve assignment data. It will then make an estimation of how much time it will take to complete each task using AI-based techniques and generate a study schedule that is optimized for the user.

Problem Definition:

As a student it can be hard to plan out your schedule in an efficient manner since assignments are spread across many courses and can become overwhelming. While Canvas does display due dates, it does not give an estimate how long an assignment will take or assist the student with distributing work across a reasonable time frame. Many students including myself will underestimate long assignments and push them to the very last minute which can lead to stress and missed deadlines. The goal of this project is to design a system that can transform a student's deadlines into a structured personalized work plan. Also I would like the system to work as autonomously as possible so that the user does not need to spend a lot of time planning their own schedule.

Proposed System (High Level):

This prototype will use the Canvas API to pull assignment data such as the title, description, point value, due date and submission type. This data can then be stored on a backend database which can be used by a time estimation module that will analyze the assignment characteristics and create estimations for the time of completion and difficulty. I have some knowledge of using predictive models from my data mining class which I think will come in handy here.

Once the estimations are complete this data will be used to create a scheduling algorithm that accounts for free space within a users calendar as well as preferred study days.

The user will have the ability to adjust the settings to their liking of how much time they would like to spend each day working so that the algorithm can create a schedule that best suits the user. It will also be able to split large assignments into multiple sessions. Once this algorithm is implemented the data will automatically be pushed to the users calendar (Google Calendar).

Some additional features could also include a feedback mechanism where the user can input how much time an assignment actually took them to complete compared to the estimate which would be used to improve the algorithm. Another interesting feature that could be useful is a timer that you can start for each individual assignment to keep you on track. From a technical standpoint this project will involve using API integration, database design, backend system development, predictive modeling, and scheduling algorithms.

System Design (Low Level):

I would say that my project has two main intelligence parts which are the prediction model which estimates the amount of time in hours needed to complete an assignment and also the scheduling algorithm which decides when the student should work on those assignments. I think that it is important to note the scope of this project. I am not proposing to create a complicated AI model since that would take too much time and is outside the scope of this project. However, a more feasible approach that I propose is to use my prior knowledge of linear regression from my data mining class to create a regression-based model. I think this would be appropriate for this project because the target output is a continuous numeric value being estimated hours of work which is perfect for a regression model.

For the next intelligence part of this project being the scheduling algorithm I would most likely use a greedy algorithm for deciding when assignments should be completed. At this stage the algorithm will already have input data such as assignments with deadlines, estimated time of completion for each assignment, and the available time slots based on the user schedule. A greedy algorithm would be able to make optimized decisions for every step of this process. For example if an assignment A was due tomorrow and takes 3 hours to complete and assignment

B is due in 5 days and takes 4 hours to complete a greedy approach would do the most urgent thing first. This algorithm would be able to tell the student what to work on and when.

Another portion of this project that is important is being able to pull and push data from canvas and google calendar. I will be using Python as a baseline for communicating with Canvas through its REST API to retrieve assignment data after the user authenticates via an access token or OAuth flow. This would allow me to retrieve JSON data holding fields such as assignment titles, due dates, descriptions, and point values. This data would need to be normalized and cleaned so that it can cleanly fit the structure that is needed to be imputed into the regression model and greedy algorithm. Once this data is collected it will be stored locally within a database. The next API that I would use is the Google Calendar API using oAuth 2.0 authorization to create and manage calendar events corresponding to the generated study sessions.

My project depends on storing multiple features such as assignments from canvas, processed features, time estimates, user preferences, feedback data etc. This is why it is important that I have a database that is able to store all of this information very efficiently. The database that I would like to use is Firebase since I have some experience with using it in the past and it is also easy to use and set up. It will work perfectly with storing data from Canvas since you can easily store JSON-like data. The only drawback to this database that I could find was it is a noSQL database. Instead of using tables like SQL it uses documents/collections to store data. However I don't think this should be a problem since my project is not very relational and shouldn't need complex joins.

Implementation Plan:

I think that it is important to have a strong step by step implementation for this project. I would probably break this up into different phases with testing in between each phase and work in a linear fashion.

Phase 1: This phase would be focused on retrieving the assignment data from the Canvas API which would authenticate the user and send requests to obtain information such as title, description, due date etc. This data will be processed and stored within the Firebase Firestore. This phase will ensure that the system has access to all the academic data it will need for the next phases of the project.

Phase 2: The purpose of this phase will be to structure the database to store all of the relevant information. This information will include the academic data that was retrieved from phase 1 and also be configured to store additional data such as user feedback, user preferences, and future model outputs. This step sets up the database to have a persistent storage system that is ready to be used for future implementations.

Phase 3: The goal of this phase is to get the time estimation component working. First I will use a rule-based technique based on certain categories such as assignment type and categories. Then I will implement a regression model to improve predictions over time. Essentially this model will take assignment features as input and output the estimated completion time.

Phase 4: Once the time estimates are available then I can start working on the scheduling algorithm to develop a study plan for the user. Here a greedy algorithm will be implemented to prioritize assignments based on urgency and also the estimated workloads of the user. Once this phase is completed the core functionality of the system will be done.

Phase 5: After generating the schedule the system will export it to the Google Calendar using the Google Calendar API. Each event will be assigned with a start time, end time, and assignment label. This last phase delivers the results in a user-friendly way ultimately finishing the project.

Potential Challenges/Risks:

There are some potential challenges that can arise from this project. My first challenge would probably be finding the data that can be used to train my regression model. To make this

work I would probably need to log how long it takes to complete assignments myself or survey other students for this information.

Another challenge/risk is the amount of competition in this area with student planners or schedulers. From the research that I have done there are a lot of schedulers out there, however they lack the automation component that my project seeks to provide.

I also question if the scope of this project is too large. This project involves using a database, multiple API's, minor machine learning knowledge, and a lot of training data. Possibly this project could be narrowed down or be fine tuned in a certain area. Also since there is so much data that needs to be stored on Firebase I could run into issues with storage limitations possibly needing to buy into a paid plan.

It is possible that I may run into certain challenges regarding authentication issues when implementing this system with Canvas and Google calendar. This is not a huge issue, just more of a time commitment to make sure that users are able to connect their accounts correctly.

Prior Art

There are several applications that were developed specifically to help students with academic planning and workload management. An example of this is Shovel which very similarly works with Canvas and other Learning Management Systems (LMS's) to import assignment data. It allows the user to input assignments, estimate the time required to complete, and generate a study schedule. Shovel provides features like workload visualization and manual time allocation which helps students to plan out their schedules.

However, tools like Shovel rely heavily on user input which does not provide a high level of automation. Students are required to manually estimate the time it takes to complete their assignments which can lead to many inaccuracies and reduce the usability of the academic planner. It also doesn't automatically pull or display all relevant assignment metadata or even automatically schedule events, this all needs to be done manually.

There are also other academic planners such as MyStudyLife and Power Planner which also allow students to track their assignments and exams in one place. However these applications all have a common theme of manual planning and estimation which don't provide optimized study plans.

Another application that is very similar is called motion which has an AI assistant which does automatically create a schedule for you. This is one of the closest applications to what I propose to build, however it still requires you to manually input the amount of time it takes you to complete certain tasks. Studies show that people consistently underestimate time and can be biased by optimism.

April 14th 3:30pm to 4:30pm